

QUICK REFERENCE FOR  
HEALTHCARE PROVIDERS

# Management of Dengue in Children

(Second Edition)



Ministry of Health  
Malaysia

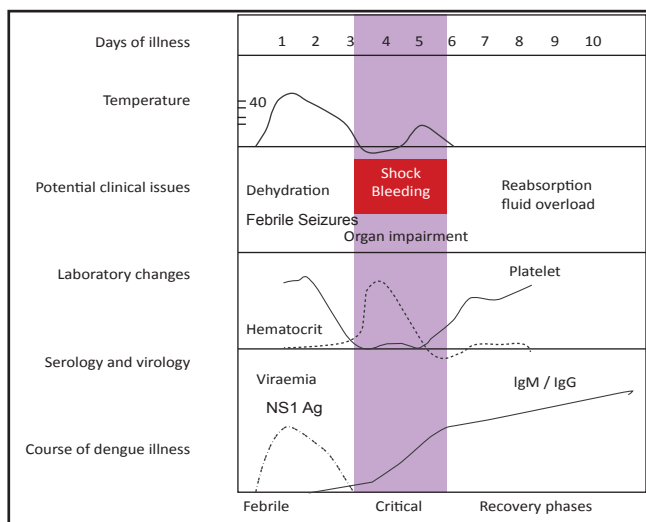


Academy of  
Medicine Malaysia

## KEY MESSAGES

1. Dengue in children has a wide spectrum of clinical presentations ranging from non-severe to life-threatening.
2. Children suspected of dengue infection should be tested with a combination of NS1 Antigen/IgM/IgG rapid test (dengue rapid combo test). ELISA for dengue may be used in centres where combo test is not offered.
3. Notification should be done for all suspected dengue cases from private and public health facilities by telephone/fax/e-notification to the nearest health office within 24 hours of diagnosis. This should be followed by written notification using the standard notification form.
4. Critical phase in dengue fever usually begins after the third (or earlier) day of illness (typically around the time of defervescence i.e. when temperature drops to and remains below 38°C). Plasma leakage may occur as a result of increased capillary permeability and is manifested by warning signs.
5. All children with dengue infection treated as outpatient should have daily clinical and laboratory monitoring using dengue record card until resolution of critical phase.
6. Isotonic crystalloid solutions should be used in resuscitation and maintenance therapy in children with dengue. Colloid solutions may be used in persistent shock despite resuscitation with crystalloid solutions.
7. Close monitoring and frequent reassessment should be done to guide appropriate fluid management of children with dengue shock.
8. Those with prolonged and/or decompensated shock should be admitted to the high-dependency or intensive care unit.
9. Blood transfusion should be given in life-threatening conditions and given as soon as severe bleeding is recognised (overt) or suspected (occult) in children with dengue.
10. Dengue infection in infants should be managed in a hospital with paediatric services.

## THE CLINICAL COURSE OF DENGUE ILLNESS



### WARNING SIGNS OF DENGUE IN CHILDREN

- Abdominal pain - abdominal tenderness and continuous pain (not intermittent)
- Persistent vomiting - ( $\geq 2$  episodes of vomiting that amounts to fatigue or requires intravenous (IV) fluids)
- Mucosal bleed - bleeding from nose, gums, conjunctiva, vagina, gastrointestinal/respiratory/urinary tract
- Lethargy, restlessness
- Liver enlargement  $>2$  cm
- Clinical fluid accumulation - pleural effusion and ascites
- Laboratory: increase in haematocrit (HCT) concurrent with decrease in platelet - HCT raised by 20% from the baseline value with concurrent decrease in platelet count  $\leq 100 \times 10^3 /\mu\text{L}$

### RISK FACTORS FOR SEVERE DENGUE IN CHILDREN

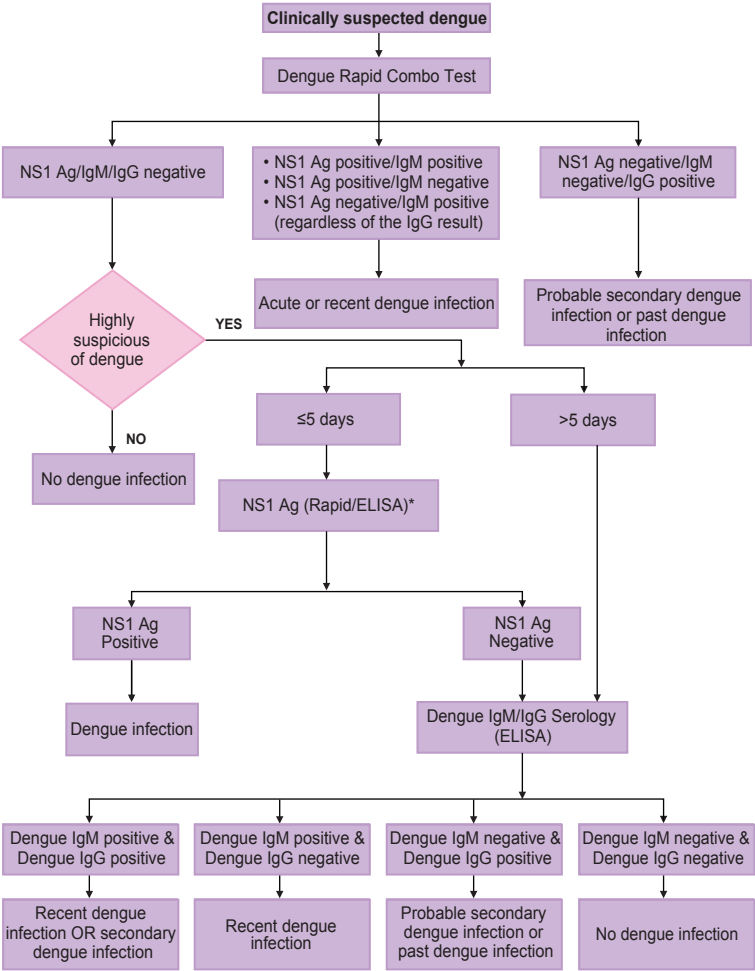
- Lethargy
- Abdominal pain
- Bleeding tendencies
- Hepatomegaly
- Haemoconcentration
- Thrombocytopenia of  $<100,000 /\mu\text{L}$

### HAEMODYNAMIC ASSESSMENT: CONTINUUM OF HAEMODYNAMIC CHANGES

| Parameters                  | Normal Circulation              | Compensated shock*                                                                                                                          | Decompensated / Hypotensive shock                 |
|-----------------------------|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| Consciousness level         | Clear and alert                 | Clear and alert                                                                                                                             | Change of mental state (restless, drowsy)         |
| Extremities                 | Warm and pink extremities       | Cold extremities                                                                                                                            | Cold, clammy extremities                          |
| Capillary refill time (CRT) | Brisk ( $<2$ sec)               | Prolonged ( $>2$ sec)                                                                                                                       | Very prolonged, mottled skin                      |
| Peripheral pulse volume     | Good volume peripheral pulses   | Weak & thready peripheral pulses                                                                                                            | Feeble or absent peripheral pulses                |
| Heart rate                  | Normal heart rate for age       | Tachycardia                                                                                                                                 | Severe tachycardia with bradycardia in late shock |
| BP                          | Normal BP for age               | <ul style="list-style-type: none"> <li>• Normal systolic pressure with raised diastolic pressure</li> <li>• Postural hypotension</li> </ul> | Hypotension/unrecordable BP                       |
| Pulse pressure              | Normal pulse pressure for age   | Narrowed pulse pressure ( $\leq 20$ mmHg)                                                                                                   | Unrecordable                                      |
| Respiratory rate            | Normal respiratory rate for age | Tachypnoea                                                                                                                                  | Metabolic acidosis/hyperpnoea                     |
| Urine output                | Normal                          | Reducing trend                                                                                                                              | Oliguria/anuria                                   |

\*unless the child is touched, parameters of shock will be missed e.g. cold extremities, weak peripheral pulses, prolonged CRT

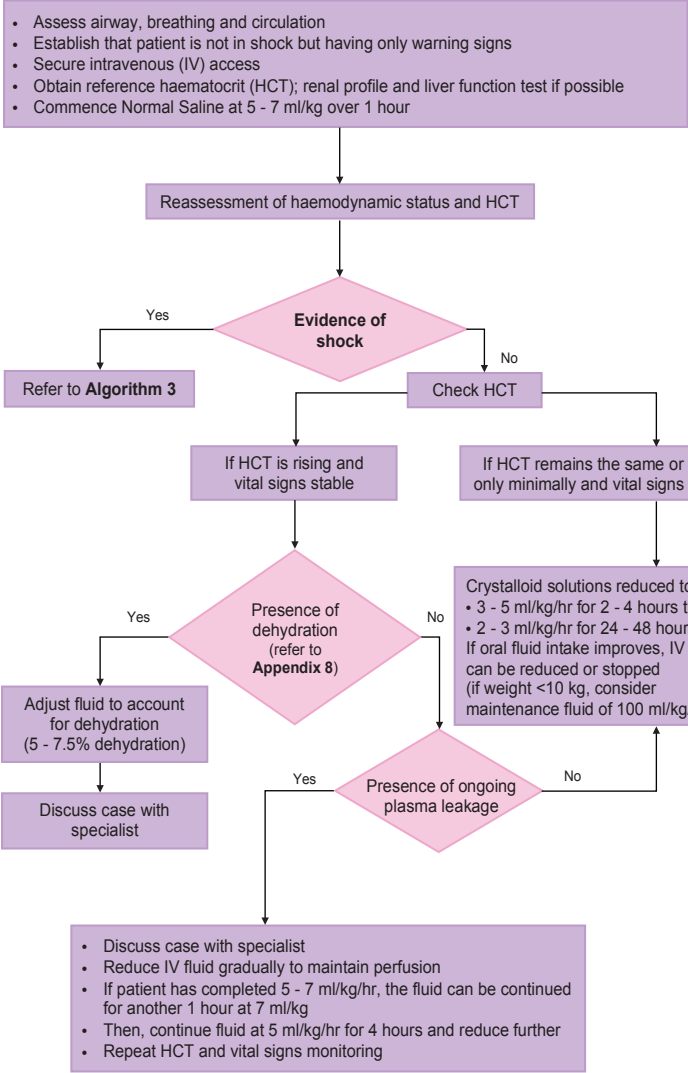
DENGUE LABORATORY DIAGNOSIS



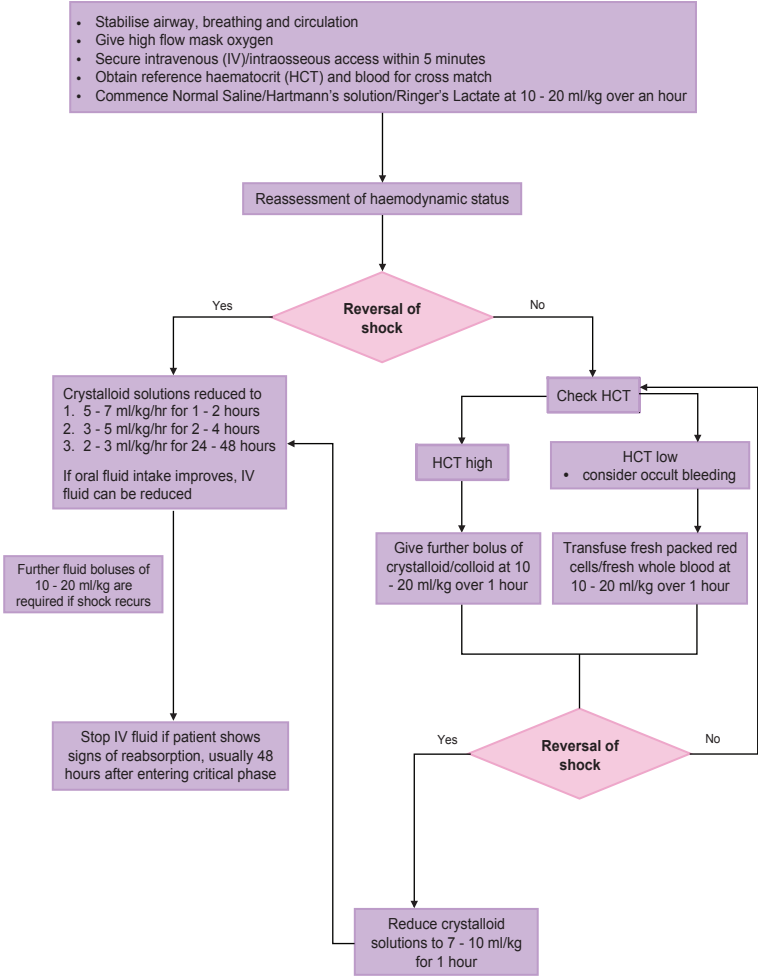
\*Alternatively, Dengue Rapid Combo Test may be repeated

- PCR should be sent if NS1 Ag/serology negative in suspected severe dengue or mortality cases.
  - ELISA for dengue may be used in centres where combo test is not offered.
- ELISA - enzyme-linked immunosorbent assay  
IgG - Immunoglobulin G  
IgM - Immunoglobulin M  
NS1 Ag - Non-structural protein 1 Antigen  
PCR - polymerase chain reaction

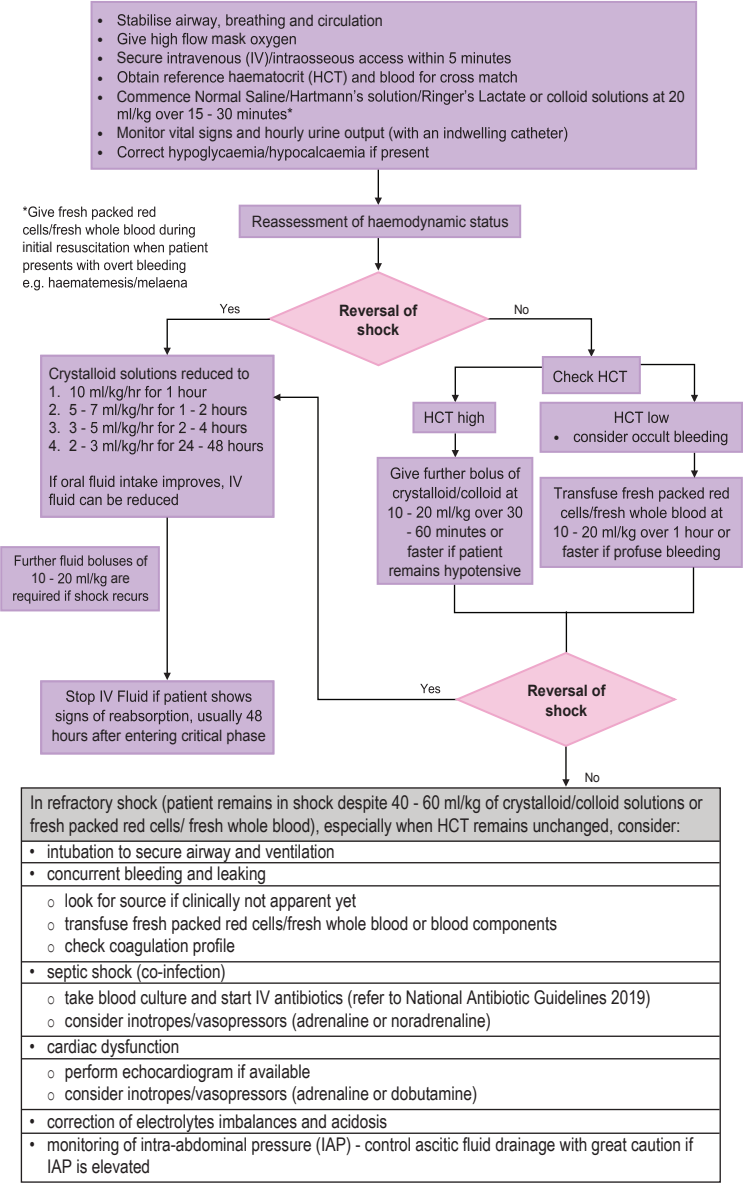
FLUID MANAGEMENT OF CHILDREN WITH DENGUE WARNING SIGNS



FLUID MANAGEMENT OF COMPENSATED DENGUE SHOCK IN CHILDREN



FLUID MANAGEMENT OF DECOMPENSATED DENGUE SHOCK IN CHILDREN



## DISCHARGE CRITERIA

|                            |                                                                                                                                                                                                                                                                                              |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Clinical criteria</b>   | <ul style="list-style-type: none"> <li>• No fever for 24 - 48 hours</li> <li>• Improvement in clinical status (general well-being, appetite, haemodynamic status, urine output)</li> <li>• Absence of respiratory distress</li> <li>• Resolution or recovery of organ dysfunction</li> </ul> |
| <b>Laboratory criteria</b> | <ul style="list-style-type: none"> <li>• Increasing trend of platelet count</li> <li>• Stable HCT without IV fluids</li> </ul>                                                                                                                                                               |

## RANGE OF HAEMATOCRIT IN DIFFERENT AGE GROUPS

| Age                | HCT (%) |
|--------------------|---------|
| Cord blood         | 45 - 65 |
| 2 weeks            | 42 - 66 |
| 3 months           | 31 - 41 |
| 6 months - 6 years | 33 - 42 |
| 7 - 12 years       | 34 - 40 |
| Adult male         | 42 - 52 |
| Adult female       | 37 - 47 |

This Quick Reference provides key messages & a summary of the main recommendations in the Clinical Practice Guidelines (CPG) Management of Dengue in Children (Second Edition)

Details of the evidence supporting these recommendations can be found in the above CPG, available on the following websites:

Ministry of Health Malaysia: [www.moh.gov.my](http://www.moh.gov.my)

Academy of Medicine Malaysia: [www.acadmed.org.my](http://www.acadmed.org.my)

### CLINICAL PRACTICE GUIDELINES SECRETARIAT

Malaysian Health Technology Assessment Section (MaHTAS)

Medical Development Division, Ministry of Health Malaysia

Level 4, Block E1, Precint 1,

Federal Government Administrative Centre 62590

Putrajaya, Malaysia

Tel: 603-88831228

E-mail: [htamalaysia@moh.gov.my](mailto:htamalaysia@moh.gov.my)